

What Drives the Ride? Assessing Factors of Immersion and Emotion in Virtual Reality Bicycle Simulations

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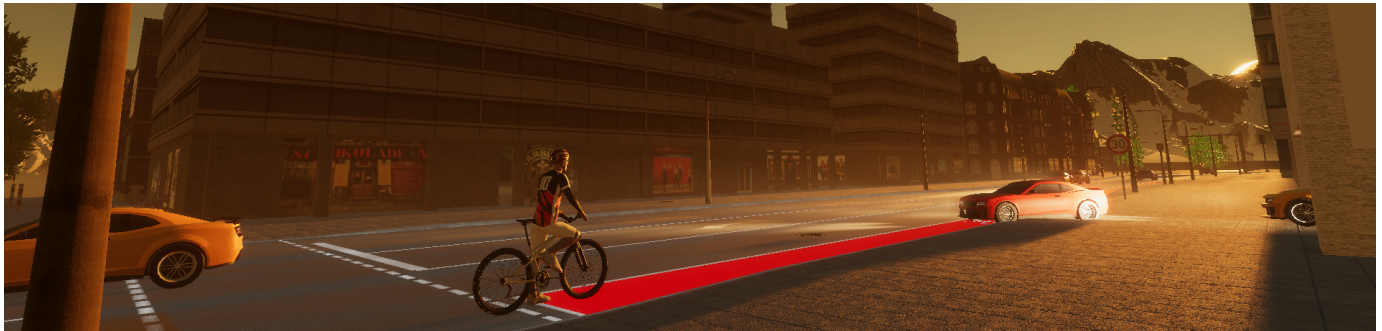


Fig. 1: The bicycle simulator enables cyclists to navigate virtual environments in a fully immersive way.

Abstract—Urban cycling offers significant environmental and mobility benefits, but is hindered by safety concerns and stress associated with traffic interactions. This study explores the use of a virtual reality bicycle simulator as a controlled environment to study cyclist behavior and improve cycling safety. Key findings indicate that participants gradually become more comfortable with the simulator, leading to reduced stress levels and enhanced immersion. Notable improvements were made to the brake, steering and acceleration systems, and the effects of these enhancements on user experience were measured in the study. Although motion sickness remains a challenge, future work includes expanding virtual environments and improving sensor accuracy. Additionally, designs will be tailored to diverse user needs, including those with a fear of cycling, to enhance immersion.

Index Terms—bicycle simulation, virtual reality, immersion

I. INTRODUCTION

The bicycle is a healthy, flexible and eco-friendly mode of transportation in urban areas [1], [2]. Despite its various advantages, many cyclists are deterred by safety concerns, as

shown in a study by DEKRA e.V. [3]. According to this study, 36 % of bike owners claim to feel (very) unsafe to ride a bike in urban areas, while 48.5 % of the interviewees cite insufficient infrastructure for bicycles such as bike lanes as the reason why they do not choose the bicycle as mode of transportation.

This lack of infrastructure not only discourages cycling but also contributes to tensions between different road users. A study by YouGov and Statista from 2024 [4] shows that 31 % of car drivers perceive cyclists as reckless, while cyclists regard cars as the greatest threat in traffic.

Recent accident statistics [5] underscore the need for further safety improvements for cyclists: Although the number of cyclists fatally injured in 2023 decreased, the total number of injuries due to bicycle accidents in traffic remains at 94,514. To address these challenges and evaluate cyclist safety in a controlled setting, bicycle simulation emerges as a promising method.

This research focuses on developing an advanced VR-based bicycle simulator, integrating sensors and hardware to connect virtual reality with real-world cycling dynamics. By allowing safe replication of rare and controlled traffic scenarios, such simulators provide valuable insights into cyclist behavior and interactions. Researchers can analyze high-risk or exceptional situations without exposing participants or others to danger, creating opportunities to design safer urban environments [6].

To meet the needs explained above, the following research questions are addressed in this paper.

- 1) How do test subjects assess the degree of immersion provided by individual sensors and actuators in the simulator?
- 2) How can sensors and actuators be modified or added to the simulator to increase the level of immersion? What options are available to enhance immersion?
- 3) How intense is the perceived level of stress for the test subjects in the simulator? Do they have a sense of control over the situation? Does the level of stress change with experience?

To address these questions, Chapter 2 provides an overview of the state-of-the-art in VR bicycle simulation. Chapter 3 introduces the bicycle simulator, which is further enhanced as detailed in Chapter 4. The methodology of the study is outlined in Chapter 5, with the results presented in Chapter 6. Finally, Chapter 7 discusses the key findings, limitations, and potential directions for future work.

II. STATE OF THE ART

This section presents an overview of the relevant literature on bicycle simulators.

Previous bicycle simulator projects have focused on studying traffic scenarios and analyzing cyclists' reactions under different conditions. The bicycle simulator utilized in this research builds upon the work of Flaig and Imschweiler [7], who developed a VR-based bicycle simulator to investigate cyclist behavior in traffic scenarios.

The tables below compare existing bicycle simulator setups that incorporate a Head-Mounted Display (HMD). Features such as components, sensors, and actuators are listed, with an "X" marking each feature explicitly mentioned in the respective study.

TABLE I: Comparison of Bicycle Simulators (Part 1)

Feature	HHN	[8]	[9]	[10]	[11]	[12]	[13]
Rocker Plate	X						
Acceleration S	X	X	X	X	X	X	X
Brake S	X	X		X	X		
Steering S	X	X	X	X	X	X	X
Roll Angle S	X	X					
Reverse S	X						
Wind A	X		X				
Height A	X						
Resistance A	X	X					X
Handlebar A					X		
Trepidation A							

TABLE II: Comparison of Bicycle Simulators (Part 2)

Feature	[14]	[15]	[16]	[17]	[18]	[19]	[20]
Rocker Plate							X
Acceleration S	X	X	X	X	X	X	X
Brake S		X	X	X	X		X
Steering S	X	X	X	X	X	X	X
Roll Angle S					X		X
Reverse S							
Wind A				X	X	X	X
Height A							
Resistance A			X				X
Handlebar A							X
Trepidation A					X		

A review of both Table 1 and Table 2 reveals that all simulators share acceleration and steering as common features, reflecting the fundamental movements in cycling. Other sensors (S) and actuators (A) are incorporated selectively, depending on the specific objectives of each research project.

Three simulators incorporate features that are not present in the Heilbronn University setup. One includes a tilt sensor, allowing users to steer using their body weight, as well as a trepidation actuator for simulating rough terrain [18]. Another simulator provides additional haptic feedback through the handlebars [11], while a third also includes a tilt sensor [8]. In contrast, the simulator developed and utilized at Heilbronn University offers three unique features: a reverse sensor, a rocker plate, and a height actuator.

III. INITIAL VR BICYCLE SIMULATOR SETUP

The scope of the research involves the improvement of certain components of the VR bicycle set-up. The study contains a comparative analysis of the performance between the initial setup and the improved version. This chapter provides an overview of the components, as well as a description of their functionalities and communication mechanisms that interconnect them.

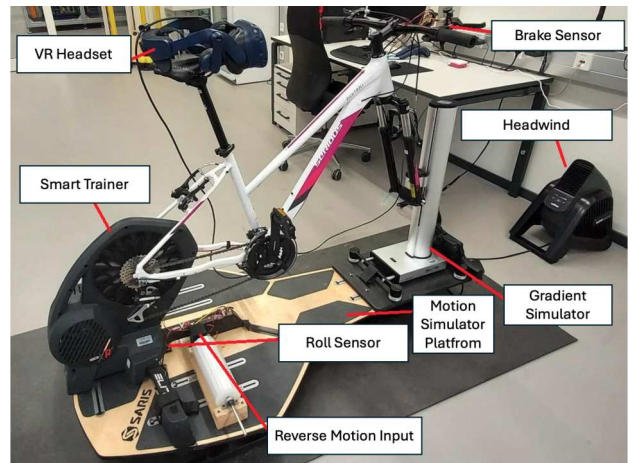


Fig. 2: VR Bicycle Simulator Setup

The initial setup of the VR bicycle simulator is shown in Figure 2. This image shows the setup in the initial configuration. Each labeled component is listed in Table III, indicating

whether it acts as a sensor or an actuator and whether it is a third-party product.

A. Smart Trainer

The Direto XR-T Smart Trainer is utilized for two primary functions: measuring pedaling speed and adjusting pedaling resistance based on the gradient elevation in the simulation. According to the official documentation [21], the Direto XR-T is designed to simulate slopes up to 24% by changing the resistance to simulate climbs and descents.

Additionally, the Direto XR-T offers a Bluetooth Low Energy (BLE) interface that enables connectivity to other home trainer components and devices. This interface is used to establish communication between the PC and Direto XR-T through a Python-based script that utilizes the Bleak library [26]. In order to read the current pedal speed, a subscription to a specific characteristic with a notify attribute is utilized, whereas the pedaling resistance is adjusted through setting the value of a certain characteristic with write attribute.

B. Gradient Simulator

To enhance the realism of rides on inclines, a ELITE Rizer gradient simulator is utilized as a front wheel mount. It is capable of adjusting the height of front wheel to inclines up to 20% and to declines down to -10% [?]. In addition, the Elite Rizer offers a steering mechanism that enables handlebar movement, allowing directional input. This mechanism registers three distinct handlebar positions: straight, left, and right.

Similar to the Smart Trainer (section III-A), the ELITE Rizer relies on a BLE interface for communication. Therefore, a comparable approach using a Python-based script that utilizes the Bleak library [26] script is employed to establish communication between the ELITE Rizer and the PC. This also means subscribing to a BLE characteristic with a notify attribute to continuously read the handlebar direction, and writing a characteristic with a write attribute to adjust the gradient elevation.

C. Headwind

The headwind is a part of every bicycle ride. To take this into account, the Wahoo KICKR headwind [23] is integrated into the setup to dynamically adjust the airflow. Because the device also relies on BLE for communication, another Python-based script utilizing the Bleak library [26] is implemented. As this device only functions as an actuator, the airflow will be adjusted according to the speed of the bicycle in the simulation by changing a specific BLE characteristic with a write attribute.

D. Virtual Reality Headset

To display the virtual reality environment (as shown in Figures 1 and 7 a HTC VIVE Pro (Head-Mounted Display) [24] is utilized. This device allows the view to be dynamically adjusted based on the user's head position and orientation. Consequently, the system accurately reflected movements such as looking up or down.

E. Motion Simulator Platform

The VR bicycle simulator setup constructed upon a Saris MP1 infinity motion simulator platform [25], which allows fore/aft and side-to-side movement. The main reason for this is to enable a natural cycling motion.

F. Roll Angle Sensor

Due to the Motion Simulator Platform described in section III-E, the bicycle can tilt side-to-side. A BNO055 [27] sensor which operates in a Nine-Degrees-of-Freedom (9DoF) mode, is mounted on the platform for continuous measurement of the roll angle. The Sensor is operated by an ESP32 microcontroller, which sends the roll angle via UDP to the PC over a shared wireless network. The received roll angle is sent to the running simulation and used to adjust the virtual bicycle's tilting in the simulation.

G. Brake Lever

Brakes are an indispensable component of every bicycle; therefore, a custom brake sensor is implemented for this setup. The rotational axis of the bicycle's brake lever is connected to a Hall-effect rotary sensor, enabling precise detection of the lever's angular position at any time. The brake lever axis is connected to the sensor axis with a coupling, while the sensor is mounted by a custom-designed 3D-printed part. The Hall-effect sensor is interfaced with an ESP32 microcontroller, which transmits brake lever data to the PC via UDP over the wireless network.

H. Reverse Motion Input

This component utilizes a bearing-mounted roll, a magnetic AS5600 encoder, and an ESP32 microcontroller. Positioned just behind the right bicycle pedal, it is easily accessible with the user's right foot. If the roll is rotated by foot, the magnetic encoder detects the rotation. As the ESP32 microcontroller continuously sends the input via UDP to the PC, simulation is able to trigger reverse motion for the virtual bicycle. This functionality is particularly beneficial in scenarios where the user encounters obstacles, providing a means to navigate out of otherwise inescapable situations within the simulation.

I. Data Flow

The communication of all components of the VR bicycle simulator is described in Figure 4. The main PC is responsible for data collection, distribution and processing as well as running the Unity simulation. The data processing node is based on multiple Python-based scripts; as already mentioned, the communication depends either on BLE or UDP. The data processing node also utilizes UDP to communicate with the Unity simulation. The VR Headset communicates with Unity via a SteamVR plugin installed in Unity.

IV. ENHANCEMENTS OF BICYCLE SIMULATOR SETUP

A significant aspect of this project is enhancing the VR bicycle setup originally developed by Flaig and Imschweiler [7]. This chapter outlines the modifications made to further improve the simulator's immersion.

TABLE III: Overview of Components and Their Categories: Sensors, Actuators, and Third-Party Devices

Component	Sensor	Actuator	Third Party
Smart Trainer	Pedaling Speed	Pedaling Resistance	ELITE DIRETO XR-T [21]
Gradient Simulator	Directional Steering	Elevation Gradient	ELITE RIZER [22]
Headwind	-	Air Resistance	Wahoo KICKR HEADWIND [23]
Virtual Reality Headset	Head Tracking	Head-Mounted Display	HTC VIVE Pro [24]
Motion Simulator Platform	-	-	Saris MP1 Platform [25]
Roll Angle Sensor	Platform Roll Angle	-	-
Brake Lever	Brake Lever Angle	-	-
Reverse Motion Input	Rotation of Roller	-	-



Fig. 3: Custom Steering Sensor

A. Steering Sensor

As described in Chapter III-B, the ELITE Rizer was used as input for the steering direction (straight, left, and right). However, due to the fact that bicycles use continuous steering, there is a significant discrepancy between the current solution and the real-life application. To drastically decrease this discrepancy, a custom steering angle sensor was added to the setup.

The new steering sensor setup employs an optical photoelectric encoder that offers two-phase directional encoding. This allows for precise angle measurement (with a resolution of 0.6 degrees per step). The encoder's axis is coupled to the bicycle fork, while the body of the encoder is mounted with a custom 3D-printed construction to the bicycle frame.

The encoder is operated by an ESP32 microcontroller, which uses interrupts for each phase to detect rotational changes of the input axis. The advantage of two phases is that, in addition to the absolute change in the rotation angle, the rotational direction can be registered. Due to small vibrations caused by the ELITE Rizer, a high-pass filter was implemented to reduce low-frequency noise. The ESP32 also relies on UDP as a communication interface with the PC.

B. Smart Trainer

Previously, the Python-based script responsible for subscribing to the pedal speed characteristic with a notify attribute and

changing the pedal resistance characteristic with a write attribute is switching between these two tasks. The modification updated this approach to ensure asynchronous functions with queues for updated values. This change allows for efficient communication and concurrent task handling.

C. Gradient Simulator

Due to the implementation of the custom steering sensor, the directional input of the ELITE Rizer is no longer required. Consequently, the Python-based scripts no longer need to subscribe to the directional steering characteristic. This reduction in communication complexity enables faster adjustment of the elevation height of the ELITE Rizer gradient simulator.

D. Brake Sensor

The brake is in need of a modification for two reasons. The first is that there is a time lag between pulling the brake lever and the PC receiving the updated value. The second reason is that the brake lever does not offer realistic resistance because there is no counterforce due to the missing rim.

To reduce the time lag, the script running on the microcontroller was adjusted to immediately send new sensor data to the PC via UDP. In order to make the brake feel more realistic, a custom 3D-printed part (see Figure 5) was added to function as replacement for the missing bicycle rim.

V. METHODOLOGY

This chapter describes how different questionnaires were used to measure the effect of the enhancements described in Chapter 4.

A. Study Design

The study consists of two parts: *Sensor Evaluation* and *Traffic Scenarios*, as illustrated in Figure 6. A crossover design was employed in both parts of the study to minimize carryover effects, ensuring that any potential influence of previous treatments on subsequent conditions was reduced [28]. This design allows each participant to experience all conditions in a counterbalanced order, helping to control for order effects and individual differences in response [28]. In this case, the sequence of treatments consisted of test rides, which

Fig. 4: Data Flow Diagram

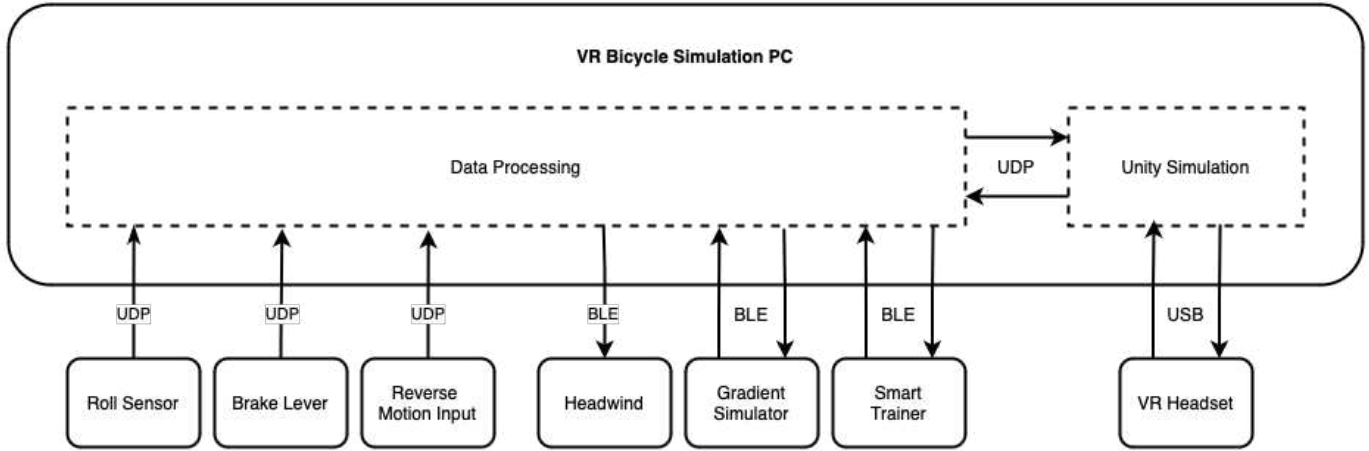


Fig. 5: Brake Rim Replacement

were alternated for each participant. This alternation helped to minimize carryover effects, allowing for a more accurate assessment of each condition.

In the *Sensor Evaluation*, participants alternated between old and new sensor configurations to evaluate their perceptions of the different sensor setups. The *Traffic Scenarios* were designed to evaluate participants' levels of stress and immersion in different simulated conditions. In both *Traffic Scenarios*, obstacles or car drivers obstructed the cyclist's use of the bicycle lane. Additionally, the conditions in both scenarios were made more challenging, either with a sunset setting or at nighttime with dense fog.

The structure of the study is outlined in Figure 6.

The questionnaires referenced in the diagram are explained in greater detail later in this chapter. Specifically:

"Motion Sickness" is assessed using the CSQ-VR questionnaire. "Immersion" is evaluated as part of the IPQ (Igroup Presence Questionnaire). Study Part 2 involves conducting a questionnaire focused on the PASA stress index. Similar to Part 1, the questions are designed to reflect on the participant's

most recent VR bicycle ride, ensuring the responses accurately capture their experience.

Since both Part 1 and Part 2 involved repeating questions twice for similar test runs, a high degree of comparability is ensured.

B. Virtual Scenarios

To evaluate the sensors and test various traffic scenarios, three distinct virtual scenarios were created. The first scenario is shown in Figure 7.

The hilly scenario (Figure 7, left) was used in both rides of Part 1. This approach was chosen to allow participants to compare two different sensor settings in the same environment. Key features of this virtual scenario include curves and hills, ensuring that all sensors and actuators are utilized. Additionally, a virtual sign prompted participants to test the brakes, as this sensor was part of the *Sensor Perception Questionnaire*.

Part 2 of the study consisted of two *Traffic Scenarios*, both intended to pose a feasible challenge to the test person. The scenarios are designed in an urban environment that resembles a German city.

Traffic Scenario A is depicted in Figure 7 (middle). It features nighttime lighting, dense fog, a construction site, and oncoming cars.

Traffic Scenario B occurs in a similar environment, this time under sunset lighting (Figure 7, right). Cars exit a parking garage and assert the right of way, while programmed scripts control the car movements to ensure they brake for the cyclist.

In both *Traffic Scenarios*, the cyclists encounter challenging situations that demand thoughtful decision-making. This setup enables precise questioning about the cyclists' stress levels and their underlying thought processes.

In summary, all virtual scenarios were carefully designed to prepare participants for the subsequent questions while presenting them with a balanced level of challenge.

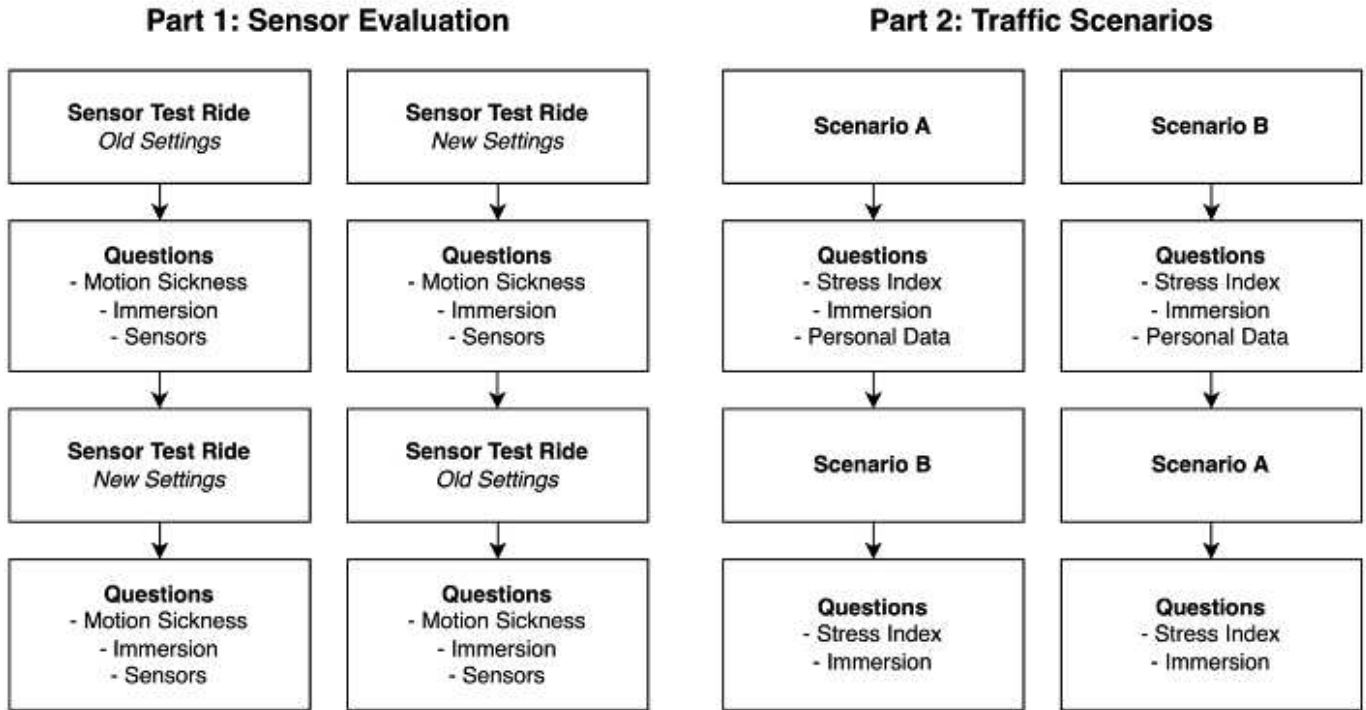


Fig. 6: Overview of the Study Design: Sensor Evaluation and Traffic Scenarios - Every participant takes either the path on the left or on the right, while both paths have equal distribution. This mitigates the carryover effect.



Fig. 7: VR Bicycle Simulation Scenarios: Sensor Evaluation (left), Traffic Scenario A (middle), and Traffic Scenario B (right)

C. Igroup Presence Questionnaire (IPQ)

The Igroup Presence Questionnaire (IPQ) is a validated tool for assessing the sense of presence experienced within a virtual environment (VE). [29] It consists of three subscales and an additional general item that does not belong to any specific subscale. The subscales are as follows:

- **Spatial Presence:** Evaluates the sensation of physically being present within the VE.
- **Involvement:** Measures the degree of attention directed toward the VE and the level of engagement experienced.
- **Experienced Realism:** Assesses the subjective perception of realism in the VE.

The general item evaluates the overall "sense of being there" and demonstrates strong correlations with all three subscales. [29]

In this study, a condensed version of the IPQ was utilized.

Each of the three subscales was assessed with a single question, and the general item was included in every instance. This condensed IPQ was administered four times: for the old sensor settings, the new sensor settings, and both *Traffic Scenarios*.

The repeated application of the IPQ emphasized immersion as the central focus of the study, recognizing it as arguably the most critical factor in VR environments. At the same time, the shortened format ensured that participants could provide their assessments efficiently, minimizing the time required for each evaluation.

D. Cybersickness in Virtual Reality Questionnaire (CSQ-VR)

The CyberSickness in Virtual Reality Questionnaire (CSQ-VR) is a standardized instrument developed to assess the symptoms of cybersickness, a condition commonly experienced by users of virtual reality (VR) systems. [30] Similar to motion sickness, cybersickness is characterized by symptoms

such as nausea, dizziness, headaches, and disorientation, which result from a sensory mismatch between the virtual environment and physical reality. The CSQ-VR quantifies the severity of these symptoms, allowing researchers and developers to evaluate user experiences in VR settings. By collecting user feedback on both physical and psychological responses to VR, the questionnaire aids in the creation of more immersive and comfortable virtual environments, with the aim of reducing cybersickness and enhancing VR usability.

The CSQ-VR comprises six questions and was administered exclusively in Part 1 to assess the impact of sensor settings on participants' well-being. Due to the brief duration of the *Traffic Scenarios* in Part 2, the CSQ-VR was not utilized in this phase.

E. Primary Appraisal Secondary Appraisal (PASA)

The PASA (Primary and Secondary Appraisal) questionnaire is a psychometric tool designed to assess situation-specific cognitive appraisals. [31] It is based on the transactional stress theory by Lazarus and Folkman [32], which distinguishes two aspects of appraisal during stress responses:

- 1) **Primary Appraisal:** Evaluates the situational circumstances for their potential threat or challenge.
- 2) **Secondary Appraisal:** Assesses the perceived coping resources and capabilities.

The PASA captures these appraisal processes through four distinct subscales. [31] For **Primary Appraisal**, the subscales include "Threat" and "Challenge," which evaluate situational factors that may be perceived as either threatening or motivating. For **Secondary Appraisal**, the subscales focus on "Self-concept of one's abilities" and "Control beliefs," assessing the individual's perceived ability to cope and exert control over the situation.

Each of these appraisals is measured using a 6-point Likert scale, providing detailed insights into participants' cognitive evaluations and stress responses. [31] This scaling approach ensures a nuanced understanding of how individuals perceive and react to specific situations, such as those encountered in the VR bicycle simulator. An individual's overall stress level ("stress index") is calculated as the difference between the primary and secondary appraisal scores. The PASA is particularly suitable for evaluating subjective stress in response to specific situations, such as a virtual reality (VR) cycling simulation.

In this study, the PASA questionnaire was utilized to evaluate participants' cognitive appraisals of stress during their interactions with the VR bicycle simulator, particularly in the context of *Traffic Scenarios A* and *B*. The PASA provided insights into how participants perceived the challenges presented by the scenarios and their ability to cope with them. To accommodate the time constraints of the study, the PASA was streamlined from its original 16 questions to 4, with a single representative question selected for each subscale.

By analyzing the development of the assessments between the two scenarios, the study aimed to identify how the experience with the simulation influenced participants' stress levels.

This approach helped to contextualize the stress index derived from the PASA and provided a comprehensive understanding of participants' subjective experiences and learning curves in the simulator.

F. Sensor Perception Questionnaire

To evaluate the current state of the sensors and to quantify the impact of the enhancements described in Chapter 4, a tailored questionnaire was developed. This questionnaire was designed to capture participants' assessments of the brake, acceleration, and steering sensors, with responses provided on a 5-point Likert scale, ranging from "not realistic at all" to "extremely realistic".

In addition to these scaled responses, the questionnaire included open-ended sections, allowing participants to elaborate on their experiences. These free-text fields enabled participants to highlight positive aspects of the simulation, identify potential drawbacks, and suggest areas for improvement. This mixed-method approach ensured a comprehensive evaluation, combining quantitative data on sensor performance with qualitative feedback to inform further refinements of the simulation system.

VI. RESULTS

This chapter presents an analysis of the study results outlined in Chapter 5. It begins with an overview of participant demographics, including their prior experience with virtual reality (VR) and cycling, and their verbalized thoughts during the study sessions. Subsequently, the findings from the questionnaires on Sensor Evaluation, Motion Sickness, Immersion, and the Stress Index are systematically examined.

A. Participants

Among the 24 participants, 17 were male and 7 were female, with a predominantly younger cohort as 20 of the participants were under 35 years old at the time of the study. The participants exhibited varying levels of familiarity with VR, with seven having no prior experience, twelve being moderately familiar with less than ten hours of usage, and four reporting extensive experience exceeding ten hours.

In terms of general cycling habits, the majority of participants (14) reported biking less than once per week, while three biked about once per week, and seven cycled several times weekly. When focusing on urban cycling, the experience was similarly skewed towards limited familiarity, with sixteen participants having minimal exposure, two reporting moderate experience, and six being highly experienced in navigating urban environments. Furthermore, the overall distribution indicates a diverse range of prior exposure to VR and cycling, providing a solid data foundation for analysis.

B. Sensor Evaluation

Feedback about the sensors was gathered using the questionnaire detailed in Section 5.6, providing structured insights into the perceived improvements and their impact on the simulation experience. The enhancements outlined in Chapter 4 were recognized by participants, as demonstrated in Figure 8.

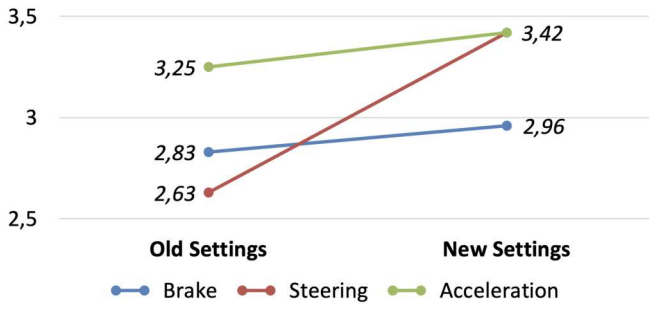


Fig. 8: Development of Sensor Evaluation

The average ratings for all three sensors clustered around a value of 3, corresponding to "moderately realistic." While the brake and acceleration logic experienced a slight increase in their ratings, the steering improvement showed a significant jump, with its average rating rising from 2.63 to 3.42.

Additional feedback from free-text responses corroborated this trend. When asked which sensors felt unrealistic, 9 participants identified the old steering system as problematic. In contrast, only three participants expressed concerns about the realism of the new steering, highlighting the noticeable improvement in this aspect of the simulation.

C. Degree of Immersion

Immersion was assessed after each of the four rides in the study, providing a substantial dataset for analysis. The progression of participants' immersion levels throughout the study is visualized in Figure 9, offering insights into how their engagement with the virtual environment evolved over time.

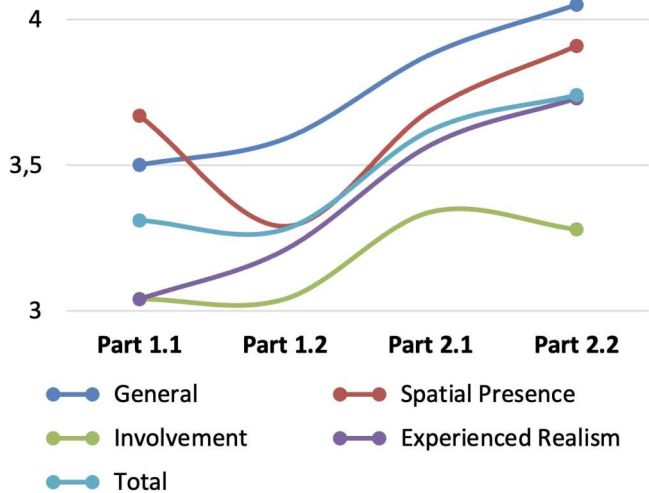


Fig. 9: Development of IPQ Score

The results indicate a positive trend in the ratings across all categories, with scores ranging from 3 to 4 out of 5, or slightly above 4, suggesting a high level of immersion. When comparing the initial value (Part 1.1) to the final value (Part 2.2), an increase in ratings is observed in all four categories.

This increase contributes to a higher overall rating, which, when averaged across the four values, shows an improvement. This suggests that the participants' experiences or perceptions have become more favorable over time, reflecting an enhanced sense of immersion.

D. Motion Sickness

An analysis of motion sickness ratings was conducted to track any changes between the two rides. Figure 10 illustrates the changes in motion sickness assessments from the first to the second ride. This assessment shows a slight decrease in motion sickness during the second ride, regardless of whether the participant tested the new or the old settings first.

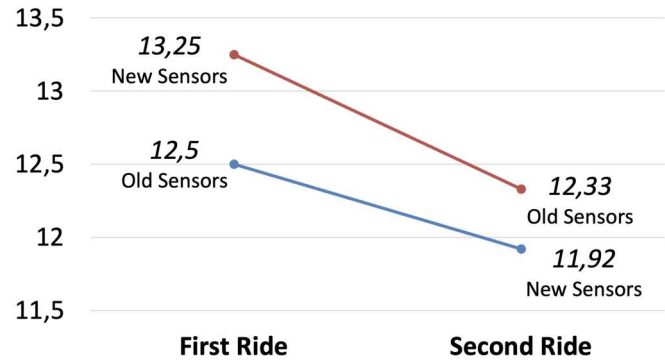


Fig. 10: Development of CSQ-VR Score

When breaking down the overall CSQ-VR score into its three components, we observe that the averages for the categories "Nausea, Dizziness" and "Disorientation, Imbalance" range between 4.5 and 5 points. A score of 5 points indicates an "intense feeling," suggesting a relatively high level of motion sickness in these areas. The "Fatigue, Discomfort" category was rated at 3, indicating a "mild feeling."

In summary, while motion sickness remains a concern with the current bicycle simulator setup, it appears to lessen with repeated experience.

E. Stress Index

In Part 2 of the study, the stress level of participants was captured with a shortened form of the PASA questionnaire. The development between both rides is displayed in Figure 11.

Regardless of whether participants started with Traffic Scenario A (Night) or Traffic Scenario B (Sunset), the stress level began at a negative value and significantly decreased in the second ride. The negative values suggest that participants were not particularly stressed initially, and the decrease indicates that their stress level dropped even further. Similar to the overall stress index, both categories of Primary Appraisal—"Threat" and "Challenge"—showed a decrease. Meanwhile, Secondary Appraisal ratings increased in both "Sense of Control" and "Self-Perception of Abilities."

The formula to calculate the stress index is as follows:

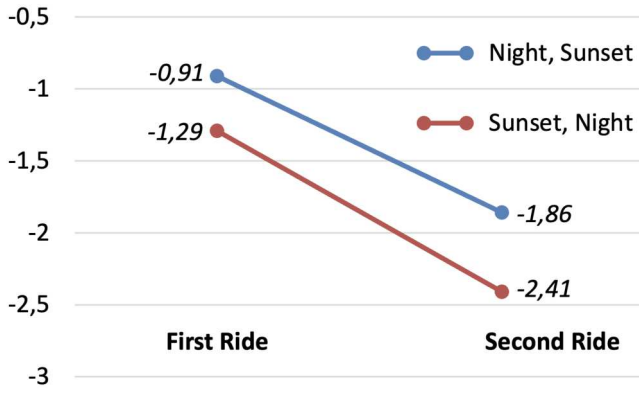


Fig. 11: Development of PASA Score

$$\text{StressIndex} = \text{PrimaryAppraisal} - \text{SecondaryAppraisal}$$

Therefore, the decrease in Primary Appraisal ratings, along with the increase in Secondary Appraisal ratings, results in a lower overall stress index. This suggests that participants became more adept at controlling the bicycle and navigating challenging situations in the simulation.

F. Thinking Aloud

The participants provided diverse feedback on the VR bicycle simulator's features during the study. Regarding sensor perception, several participants highlighted the impact of lags on the overall cycling experience. However, the progression of familiarity with the simulator was evident, as one participant remarked, "The more I did it, the more relaxed I became. I learned how the bike responds and now feel like riding even more." The simulation's acceleration and braking mechanisms received mixed reviews: one participant commented, "The speed model could be improved," while another praised the braking feature, stating, "The brake pad makes the braking feel significantly better."

In the Traffic Scenarios, the urban environment was deemed more realistic than the forest setting by several participants. Traffic Scenario A elicited empathy from one participant, who apologized to virtual drivers for perceived interruptions, while Traffic Scenario B prompted frustration, demonstrated when another cyclist reacted to a car's behavior with, "What's that supposed to mean?" These scenarios evoked real-world parallels, with a participant humorously noting, "This reminds me of Mannheim," as they blocked virtual cars.

Some participants experienced mild motion sickness during or after the simulation. The sensations were likened to "a mix between rocking and a rollercoaster," with another adding, "Like when you get off a trampoline." Specific events, such as braking, exacerbated symptoms for some, with one stating, "When I braked, I immediately felt sick again." Individual factors influenced these reactions, with one participant mentioning a prior bicycle accident and another highlighting their prolonged absence from cycling.

VII. DISCUSSION

This chapter presents conclusions, discusses limitations, and highlights future research opportunities.

A. Key Takeaways

As highlighted by the decrease in the stress index (Chapter 6.5) and the increase in immersion (Chapter 6.3), the study participants became more comfortable using the simulator. This suggests that, similar to real-world cycling, the ability to navigate the virtual environment improves with experience. This finding highlights the simulator's potential as a training tool, allowing riders to gain cycling experience in a safe and controlled environment.

Despite these positive developments, motion sickness remains an issue that needs to be addressed. Future improvements should aim to enhance both immersion and reduce motion sickness.

As discussed in Chapter 6.3, participants acknowledged improvements to the sensors, with the most significant enhancement observed in steering. However, not all the assessments showed the same degree of change, indicating some variability in how improvements were perceived. To make more targeted improvements, the suggestions outlined in Chapter 7.3 will be analyzed. With further improvements to the simulator, especially regarding immersion, the simulator could better reflect real-world cycling scenarios, making it useful for urban mobility research and training.

B. Limitations

In this study, several limitations should be acknowledged. First, the sample size was relatively small, which may limit the generalizability of the results to a broader population. All of the participants had prior experience in cycling, which could have influenced their ability to adapt to the simulator more quickly. Additionally, the simulator's settings, including the virtual environments, may not fully replicate real-world cycling conditions.

Another limitation lies in the motion sickness assessments, as individual susceptibility to motion sickness varies significantly, and the ratings may not accurately reflect the experiences of all users. The study also relied on self-reported data, which can introduce bias or inaccuracies due to the subjective nature of the assessments.

Finally, not all sensors in the simulator worked perfectly at all times, which could have led to inconsistent experiences for some participants. These limitations should be considered when interpreting the findings, and future research could address these gaps for a more comprehensive understanding.

C. Future Work

Several options have been identified to enhance the bicycle simulator. One potential improvement involves the inclusion of additional scenarios, such as those with trucks, buses, pedestrians, and varying environments with different bike lanes. Expanding the virtual city by replicating real urban

areas using specific tools or AI-driven generation of new city sections could further increase realism.

Enhancing the traffic simulation with more complex intersections, alongside a broader variety of vehicle behaviors (fast vs. slow, safe vs. dangerous), would create a more dynamic experience. The sound design could also be enriched by incorporating more immersive elements, including bicycle noises, car engines, horns, pedestrian sounds, and environmental features like wind and birds.

Adding interactive elements such as a bicycle bell and incorporating gesture and speech recognition for communication with other traffic participants would improve user engagement. Improving the braking system by adding a second brake, and providing separate control for the front and rear wheels, along with simulating suspension during braking, could make the experience more realistic. Similarly, enhancing the steering sensor system by integrating tilt values, gyroscopic forces, and speed considerations could further improve the simulation's responsiveness.

Lastly, optimizing the simulator for participants with cyclophobia by exploring their specific needs, status, and perspectives would make the simulator more accessible and inclusive. Additionally, incorporating physiological measures could provide valuable insights into how to further tailor the experience to reduce anxiety and enhance comfort for these participants. These suggestions could significantly contribute to a more realistic, immersive, and adaptable cycling experience for a wide range of users.

REFERENCES

- [1] J. Heimel, *PendlerRatD-Studie 2022: Ergebnisbericht*. Göttingen: Cuvillier Verlag, 2022.
- [2] J. Heimel and I. Balzer, "Förderung des fahrradpendelverkehrs in deutschland: Ergebnisse der nachbefragung des projekts PendlerRatD," *Internationales Verkehrswesen, Mobilität: Wissenschaft*, vol. 1, pp. 9–18, 2024.
- [3] DEKRA. (2023) Sicherheit im straßenverkehr: Rund jeder dritte fühlt sich auf dem fahrrad unsicher. Available: <https://www.dekra.de/de/sicherheit-im-strassenverkehr-rund-jeder-dritte-fuehlt-sich-auf-dem-fahrrad-unsicher>.
- [4] Statista and YouGov. (2024) YouGov-Umfrage zum straßenverkehr 2024. [Online]. Available: https://de.statista.com/presse/p/yougov_umfrage_stra_enverkehr_2024/
- [5] Statistisches Bundesamt (Destatis). (2024) Verletzte bei verkehrsunfällen nach art der verkehrsbeteiligung. [Online]. Available: <https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Verkehrsunfaelle/Tabellen/verletzte-fahrzeugart.html>
- [6] German Federal Highway Research Institute (BASt). Bicycle simulator for traffic safety research. [Online]. Available: https://www.bast.de/EN/Traffic_Safety/Technology/bicycle-simulator.html
- [7] D. Flaig, H. Imschweiler, J. Heimel, and G. Meixner, "Implementing a VR-Enabled bicycle simulator: Integrating physical sensor and actuator data for a realistic simulation experience," in *Proceedings of the 16th International Conference on Intelligent Human Computer Interaction*, 2024.
- [8] F. Schramka, S. Arisona, M. Joos, and A. Erath, "Development of virtual reality cycling simulator," Future Cities Laboratory (FCL), Singapore, Working Paper 1244, 2017.
- [9] M. Löchtefeld, A. Krüger, and H. Gellersen, "DeceptiBike-Assessing the perception of speed deception in a virtual reality training bike system," in *Proceedings of the 9th Nordic Conference on Human-Computer Interaction*, 2016, pp. 1–10.
- [10] J. Mittelstaedt, J. Wacker, and D. Stelling, "Effects of display type and motion control on cybersickness in a virtual bike simulator," *Displays*, vol. 51, pp. 43–50, 2018.
- [11] G. Hernández-Melgarejo, D. A. Flores-Hernández, A. Luviano-Juárez, L. A. Castañeda, I. Chairez, and S. D. Gennaro, "Mechatronic design and implementation of a bicycle virtual reality system," *ISA Transactions*, vol. 97, pp. 336–351, 2020.
- [12] T. von Sawitzky, P. Wintersberger, A. Löcken, A.-K. Frison, and A. Riener, "Augmentation concepts with HUDs for cyclists to improve road safety in shared spaces," in *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020, pp. 1–9.
- [13] G. Byrd, "Cycling through cyberspace," *Computer*, vol. 48, no. 8, pp. 72–75, 2015.
- [14] D. Ullmann, J. Kreimeier, T. Götzmann, and H. Kipke, "BikeVR: A virtual reality bicycle simulator towards sustainable urban space and traffic planning," in *Proceedings of the Conference on Mensch Und Computer*, 2020, pp. 511–514.
- [15] S. O'Hern, J. Oxley, and M. Stevenson, "Validation of a bicycle simulator for road safety research," *Accident Analysis & Prevention*, vol. 100, pp. 53–58, 2017.
- [16] M. Nazemi, M. A. B. van Eggermond, A. Erath, D. Schaffner, M. Joos, and K. W. Axhausen, "Studying bicyclists' perceived level of safety using a bicycle simulator combined with immersive virtual reality," *Accident Analysis & Prevention*, vol. 151, p. 105943, 2021.
- [17] M. Melo, T. Rocha, L. Barbosa, and M. Bessa, "Presence in virtual environments: Objective metrics vs. subjective metrics — a pilot study," in *2016 23rd Portuguese Meeting on Computer Graphics and Interaction*, 2016, pp. 1–6.
- [18] R. de Souza e Almeida, R. Cherullo, D. Radetic, G. F. M. Silva-Calpa, and A. B. Raposo, "Exploring motor and sensory devices in a bicycle simulator," in *2019 21st Symposium on Virtual and Augmented Reality*, 2019, pp. 232–241.
- [19] A. Hamad, "Two wheelistic: Development of a high-fidelity virtual reality cycling simulator for transportation safety research," Ph.D. dissertation, University of Michigan-Dearborn, 2021.
- [20] P. Wintersberger, A. Matvienko, A. Schweidler, and F. Michahelles, "Development and evaluation of a motion-based VR bicycle simulator," *Proceedings of the ACM on Human-Computer Interaction*, vol. 6, no. MCHI, p. 210, 2022.
- [21] Elite S.r.l. (2025) Direto XR-T product documentation. [Online]. Available: https://www.elite-it.com/uploads/product/catalog_box_cta_file_en/189/DIRETO_XR-T_2025_EN.pdf
- [22] E. S.r.l. (2025) Rizer produktdokumentation. [Online]. Available: https://www.elite-it.com/uploads/product/catalog_box_cta_file_de/201/RIZER_2025_DE.pdf
- [23] Wahoo Fitness. KICKR HEADWIND quick start guide. [Online]. Available: https://www.wahoofitness.com/media/wysiwyg/instructions/Wahoo_KICKR_HEADWIND_QSG.pdf
- [24] HTC Corporation. VIVE Pro HMD user guide. [Online]. Available: https://developer.vive.com/documents/719/VIVE_Pro_HMD_User_Guide.pdf
- [25] Saris. MP1 Nfinity technology motion platform product page. [Online]. Available: <https://saris.com/products/mp1-nfinity-technology-motion-platform>
- [26] H. Blidh. Bleak: Python library for bluetooth low energy. [Online]. Available: <https://bleak.readthedocs.io/>
- [27] Bosch Sensortec. BNO055 intelligent 9-axis absolute orientation sensor datasheet. [Online]. Available: <https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bno055-ds000.pdf>
- [28] B. Jones and M. G. Kenward, *Design and Analysis of Cross-Over Trials*, 2nd ed. New York: Chapman and Hall/CRC, 2003.
- [29] Igroup Project. Igroup presence questionnaire (IPQ) homepage. [Online]. Available: <https://www.igroup.org/pq/ipq/index.php>
- [30] P. Kourtesis, J. Linnell, R. Amir, F. Argelaguet, and S. E. MacPherson, "Cybersickness in virtual reality questionnaire (CSQ-VR): A validation and comparison against SSQ and VRSQ," *Virtual Worlds*, vol. 2, no. 1, pp. 16–35, 2023.
- [31] J. Gaab, "PASA - primary appraisal secondary appraisal - ein fragebogen zur erfassung von situationsbezogenen kognitiven bewertungen," *Verhaltenstherapie*, vol. 19, no. 2, pp. 114–115, 2009.
- [32] R. S. Lazarus and S. Folkman, *Stress, Appraisal, and Coping*. New York: Springer Publishing Company, 1984.